Setting up the database: For installing PostgreSQL, user creation, creating the database and dumping the schema, use the following commands:

```
apt-get install postgresql
     postgres createuser -P firmadyne, firmadyne
     postgres createdb -O firmadyne firmware
     postgres pssql -d firmware < ./firmadyne/database/schema
```

Installing binaries: Go to the *firmadyne* folder and execute the *download.sh* file, which downloads all the pre-built binaries:

```
cd firmadyne; ./download.sh
```

Installing Qemu for virtualisation: To install Qemu, give the following command:

```
sudo apt-get install qemu-system-arm qemu-system-mips qemu-
system-x86 qemu-utils
```

Once setup is completed, the firmware is mounted for dynamic analysis. For this demonstration, I have downloaded the *DLink DIR-822_REVB_FIRMWARE_2.03b01.BIN* file from the DLink vendor's official website.

After downloading the firmware, let us try to extract the file system of the firmware using the Python script *extractor.py*, as shown in the code below. Once the extraction process begins, a tag is assigned to the firmware. As shown in the code below, Database ID 8 is assigned to the firmware, which is being extracted. The extraction of the firmware results in a *.gz* file being created in the *images/* folder. This *.gz* file carries a number related to its database image ID.

```
root@kali:~/firmadyne# ./sources/extractor/extractor.py -b
Dlink -sql 127.0.0.1 -np -nk "DIR-822_REVB_FIRMWARE_2.03b01.
BIN" images/
>> Database Image ID: 8

/root/firmadyne/DIR-822_REVB_FIRMWARE_2.03b01.BIN
>> MD5: b666a447ebabdec1db7672048b8e6d0a
>> Tag: 8

>> Temp: /tmp/tmp2FSvWv
>> Status: Kernel: True, Rootfs: False, Do_Kernel: False,
Do_Rootfs: True
>> Recursing into archive ...
>>>> Squashfs filesystem, little endian, version 4.0,
compression:lzma, size: 5025153 bytes, 2282 inodes,
blocksize: 131072 bytes, created: 2016-03-02 07:07:37
>>>> Found Linux filesystem in /tmp/tmp2FSvWv/_DIR-822_REVB_
FIRMWARE_2.03b01.BIN.extracted/squashfs-root!
>> Skipping: completed!
>> Cleaning up /tmp/tmp2FSvWv...
```

Once extraction is completed, the architecture of the firmware is identified and the output is stored in the image table of the database with the help of the *getArch.sh* file, as shown in the code below.

```
root@kali:~/firmadyne# ./scripts/getArch.sh ./images/8.tar.gz
./bin/busybox: mipseb
Password for user firmadyne:
```

In the next step, the contents of the file system for the firmware are loaded. As mentioned earlier, the contents from *8.tar.gz,* which is located under the *images* folder, are loaded. Use the following command:

```
./scripts/tar2db.py -i 8 -f ./images/8.tar.gz
```

Next, a QEMU disk image is created, as shown in the code below.

```
root@kali:~/firmadyne# ./scripts/tar2db.py -i 8 -f ./images/8.
tar.gz

root@kali:~/firmadyne# ./scripts/makeImage.sh 8
Querying database for architecture... Password for user
firmadyne:
mipseb
----Running----
----Copying Filesystem Tarball----
----Creating QEMU Image----
Formatting '/root/firmadyne/scratch//8//image.raw', fmt=raw
size=1073741824
----Creating Partition Table----

Welcome to fdisk (util-linux 2.31.1).
Changes will remain in memory only, until you decide to write
them.
Be careful before using the write command.

Device does not contain a recognized partition table.
Created a new DOS disklabel with disk identifier 0x99029eac.

Command (m for help): Created a new DOS disklabel with disk
identifier 0x2f73ce76.

Command (m for help): Partition type
   p   primary (0 primary, 0 extended, 4 free)
   e   extended (container for logical partitions)
Select (default p): Partition number (1-4, default 1): First
sector (2048-2097151, default 2048): Last sector, +sectors or
+size{K,M,G,T,P} (2048-2097151, default 2097151):
Created a new partition 1 of type 'Linux' and of size 1023 MiB.

Command (m for help): The partition table has been altered.
Syncing disks.
```